SURVIVAL PREDICTION OF TITANIC USING CLASSIFICATION

Business Intelligence

Professor : Giovanna Maria Dimitri

Sony Kumari

Engineering Management
Department of Information engineering and mathematics
University of Siena
[2023/2024]

Abstract

This project employs various predictive modeling approach from regression and classification to dissect the demographic and social variables that potentially influenced survival rates among the Titanic passengers. Using passenger data such as names, ages, genders, and socio-economic statuses, the model aims to predict and analyze patterns of survival, shedding light on whether some groups of individuals had a higher likelihood of surviving the disaster. The objective is to enhance our understanding of historical human behavior under crisis and to contribute to the broader discourse on safety and discrimination in emergency evacuations. The insights gained could serve as a poignant reminder of the past failures and guide future strategies in managing human safety effectively.

- *Research Question* : Predicting survival on the Titanic based on passenger data
- *Algorithm Used* : Logistic Regression, Decision Tree, Naïve Bayes, Random forests and K-nearest neighbours

---

Introduction and Background

Overview of the Titanic Dataset: As a student exploring the world of data, the Titanic dataset presents an invaluable opportunity for practical learning. Hosted on Kaggle, this dataset comprises information from 891 of the 2,224 passengers and crew aboard the RMS Titanic, which tragically sank in the year1912. Key attributes in the dataset include : *survival status, passenger class, age, and fare, among others, providing a rich basis for analysis.*

Dataset Composition: The Titanic dataset is divided into two main sets:

1. **Training Set (train.csv):** This dataset is used to build machine learning models. It includes both features and the outcome (survival status) for each passenger, known as the "ground truth." The training set allows for the training of models on known outcomes to learn the patterns in the data.
2. **Test Set (test.csv):** Used to evaluate the performance of the machine learning models on unseen data. The test set includes features but does not reveal the survival outcome for the passengers. The goal is to predict these outcomes based on the model trained using the training set.

3. Additionally, there is a `gender_submission.csv` file, which serves as an example of what a submission file should look like. This file assumes that all female passengers survived, illustrating one possible method of simple prediction based on gender.

Features and Data Types: The key features provided in the dataset include:

- **PassengerId** (integer): Identifier for each traveler.
- **Survived** (integer): Represents the survival status (1 = survived, 0 = did not survive).
- **Pclass** (integer): Ticket class (1 = 1st class, 2 = 2nd class, 3 = 3rd class) as a proxy for socio-economic status.
- **Name** (string): Passenger names.
- **Sex** (string): Gender of the passengers.
- **Age** (float): Age in years.
- **SibSp** (integer): Number of siblings or spouses aboard.

- **Parch** (integer): Number of parents or children aboard.
- **Ticket** (string): Ticket number.
- **Fare** (float): Passenger fare.
- **Cabin** (string): Cabin number.
- **Embarked** (string): Port of embarkation (C = Cherbourg, Q = Queenstown, S = Southampton).
- **Training**(boolean) : TRUE for training set, FALSE for test set.

```
# check structure of training set
str(train.data)
```

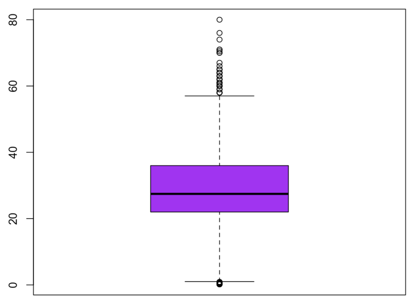
```
'data.frame':   891 obs. of  13 variables:
 $ PassengerId: int  1 2 3 4 5 6 7 8 9 10 ...
 $ Survived   : int  0 1 1 1 0 0 0 0 1 1 ...
 $ Pclass     : int  3 1 3 1 3 3 1 3 3 2 ...
 $ Name       : Factor w/ 891 levels "Abbing, Mr. Anthony",..: 109 191 358 277 16
559 520 629 417 581 ...
 $ Sex        : Factor w/ 2 levels "female","male": 2 1 1 1 2 2 2 2 1 1 ...
 $ Age        : num  22 38 26 35 35 NA 54 2 27 14 ...
 $ SibSp      : int  1 1 0 1 0 0 0 3 0 1 ...
 $ Parch      : int  0 0 0 0 0 0 0 1 2 0 ...
 $ Ticket     : Factor w/ 681 levels "110152","110413",..: 524 597 670 50 473 276
86 396 345 133 ...
 $ Fare       : num  7.25 71.28 7.92 53.1 8.05 ...
 $ Cabin      : Factor w/ 148 levels "","A10","A14",..: 1 83 1 57 1 1 131 1 1 1
...
 $ Embarked   : Factor w/ 4 levels "","C","Q","S": 4 2 4 4 4 3 4 4 4 2 ...
 $ training   : logi  TRUE TRUE TRUE TRUE TRUE TRUE ...
```

Data Preparation and Cleaning

Add a survived column in test.csv and populate it as N/A and bind it with training.csv as a one file named 'full.data' comprising both csv file ready for cleaning. Data Summary shown in the *figure in the below left hand side*. Columns 'Age' and 'Fare' have N/A value as 263 and 1 respectively, 'Cabin' has 1014 missing values, 'Ticket' has 1261 and 'Embarked' has 2 values.

- Ignoring 'Cabin' since more than 50% values are missing. For age there are outliers, as *shown in the figure in the below right hand side*.

```
  PassengerId    Survived    Pclass                              Name     
 Min.   :   1   0    :549   1:323   Connolly, Miss. Kate           :   2  
 1st Qu.: 328   1    :342   2:277   Kelly, Mr. James               :   2  
 Median : 655   NA's:418    3:709   Abbing, Mr. Anthony            :   1  
 Mean   : 655                       Abbott, Mr. Rossmore Edward    :   1  
 3rd Qu.: 982                       Abbott, Mrs. Stanton (Rosa Hunt):   1  
 Max.   :1309                       Abelson, Mr. Samuel            :   1  
                                    (Other)                        :1301  
     Sex           Age            SibSp            Parch           Ticket    
 female:466   Min.   : 0.17   Min.   :0.0000   Min.   :0.000   CA. 2343:  11  
 male  :843   1st Qu.:21.00   1st Qu.:0.0000   1st Qu.:0.000   1601    :   8  
              Median :28.00   Median :0.0000   Median :0.000   CA 2144 :   8  
              Mean   :29.88   Mean   :0.4989   Mean   :0.385   3101295 :   7  
              3rd Qu.:39.00   3rd Qu.:1.0000   3rd Qu.:0.000   347077  :   7  
              Max.   :80.00   Max.   :8.0000   Max.   :9.000   347082  :   7  
              NA's   :263                                      (Other) :1261  
      Fare                     Cabin      Embarked  training      
 Min.   :  0.000                  :1014    :  2    Mode :logical  
 1st Qu.:  7.896   C23 C25 C27    :   6   C:270    FALSE:418      
 Median : 14.454   B57 B59 B63 B66:   5   Q:123    TRUE :891      
 Mean   : 33.295   G6             :   5   S:914                   
 3rd Qu.: 31.275   B96 B98        :   4                           
 Max.   :512.329   C22 C26        :   4                           
 NA's   :1         (Other)        : 271                           
```



- Columns 'Age' and 'Fare' handled by : if there is any missing value, then replace it with predicted value by linear regression. After and Before data distribution :

Summary of 'Age' before handling data :

```
Min. 1st Qu.  Median    Mean 3rd Qu.    Max.    NA's
0.17   21.00   28.00   29.51   38.00   66.00     263
```

After handling data:

```
Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.17   22.00   27.44   29.44   36.00   80.00
```

For 'Fare' before handling data:

```
Min. 1st Qu.  Median    Mean 3rd Qu.    Max.    NA's
0.000   7.879  13.000  17.962  26.000  65.000       1
```

After handling it:

```
Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.000   7.896  14.454  33.277  31.275 512.329
```

- Handling categorical variables : Just 2 values are missing in 'Embarked' column, hence populated it with mode of the field.

Model Prediction

We'll use five different types of classification models to choose from when making our final predictions. These models are:

- **Logistic Regression**
- **Naive Bayes**
- **Decision Tree**
- **Random Forest**
- **K Nearest Neighbour**

Each model will use the same formula that we will create below :

survived_equation <- as.formula("Survived ~ Pclass + Sex + SibSp + Parch + Fare + Embarked + Age")

We need to split our cleaned 'full.data' back into their respective train and test sets.

 # create dataframe of 891 train data.

```
'data.frame':   891 obs. of  9 variables:
 $ PassengerId: int  1 2 3 4 5 6 7 8 9 10 ...
 $ Survived   : Factor w/ 2 levels "0","1": 1 2 2 2 1 1 1 1 2 2 ...
 $ Pclass     : Factor w/ 3 levels "1","2","3": 3 1 3 1 3 3 1 3 3 2 ...
 $ Sex        : Factor w/ 2 levels "female","male": 2 1 1 1 2 2 2 2 1 1 ...
 $ Age        : num  22 38 26 35 35 ...
 $ SibSp      : int  1 1 0 1 0 0 0 3 0 1 ...
 $ Parch      : int  0 0 0 0 0 0 0 1 2 0 ...
 $ Fare       : num  7.25 71.28 7.92 53.1 8.05 ...
 $ Embarked   : Factor w/ 4 levels "","C","Q","S": 4 2 4 4 4 3 4 4 4 2 ...
```

create dataframe of 418 test data

```
data.frame':   418 obs. of  8 variables:
$ PassengerId: int  892 893 894 895 896 897 898 899 900 901 ...
$ Pclass     : Factor w/ 3 levels "1","2","3": 3 3 2 3 3 3 3 2 3 3 ...
$ Sex        : Factor w/ 2 levels "female","male": 2 1 2 2 1 2 1 2 1 2 ...
$ Age        : num  34.5 47 62 27 22 14 30 26 18 21 ...
$ SibSp      : int  0 1 0 0 1 0 0 1 0 2 ...
$ Parch      : int  0 0 0 0 1 0 0 1 0 0 ...
$ Fare       : num  7.83 7 9.69 8.66 12.29 ...
$ Embarked   : Factor w/ 3 levels "C","Q","S": 2 3 2 3 3 3 2 3 1 3 ...
```

Logistic Regression :

```
log_reg <- glm(survived_equation, data = clean.training, family = "binomial")
```

Naïve Bayes :

```
bayes <- naive_bayes(survived_equation, data = clean.training)
```

Decision Tree :

```
decTree <- rpart(survived_equation, data = clean.training, method = "class")
```

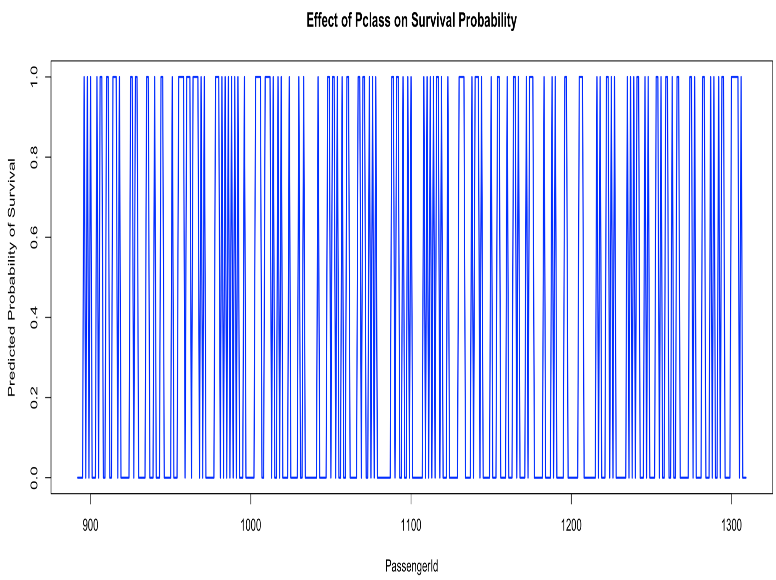
Random Forrest :

```
rand_for <- randomForest(survived_equation, data = clean.training, ntree = 500, mtry
= 3)
```

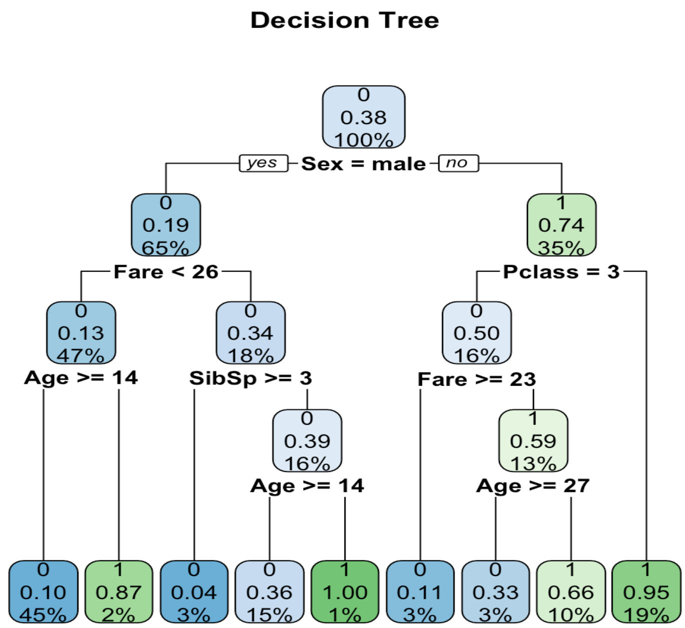
K Nearest Neighbour :

```
knn.training <- clean.training %>%
 select(Pclass, Sex, SibSp, Parch, Fare, Embarked, Age)
```

Outcome of Logistic Regression :

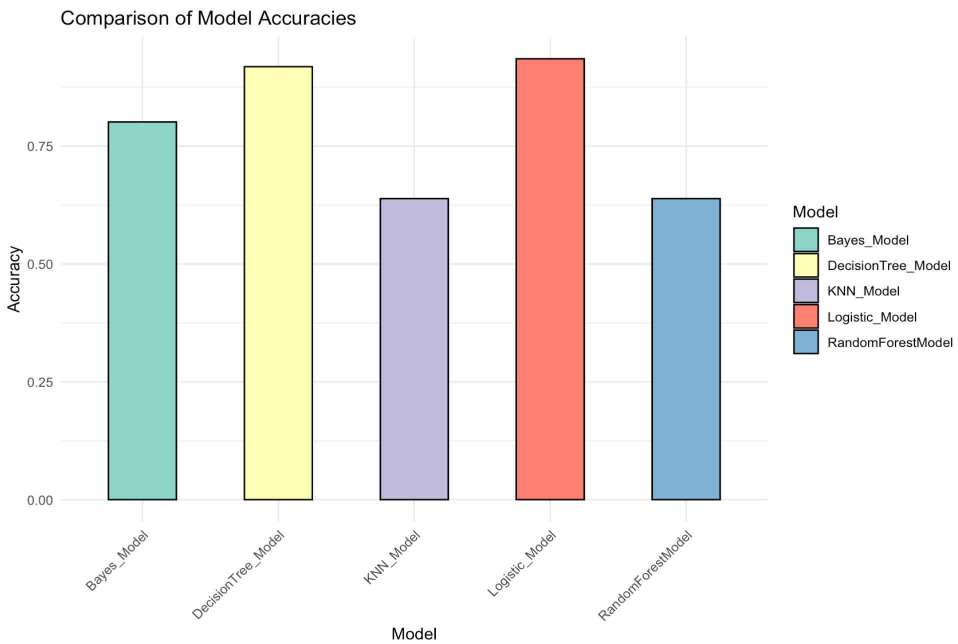


Outcome of Decision Tree :



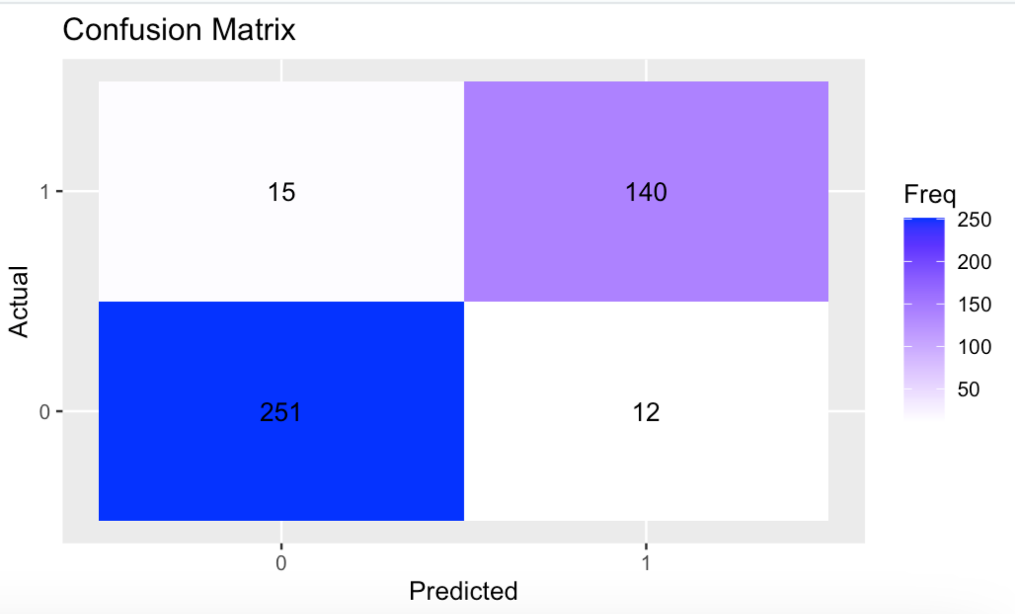
The Accuracy of the classification models with the test dataset are as follows :

- Logistic Regression: 76.32%
- Naive Bayes: 74.88%
- Decision Tree: 78.47%
- Random Forest: 77.75%
- K Nearest Neighbor: 66.03%



Apart from accuracy score we need the performance metric like ‘Precision’ and ‘Recall’ as per the type of problem which needs to be assessed. The confusion matrix thus help us with an understanding of such values.

Confusion Matrix for Logistic Regression :




```
>
> # Print recall and precision
> print(paste("Recall for class 0: ", recall[1]))
[1] "Recall for class 0:  0.943609022556391"
> print(paste("Recall for class 1: ", recall[2]))
[1] "Recall for class 1:  0.921052631578947"
> print(paste("Precision for class 0: ", precision[1]))
[1] "Precision for class 0:  0.954372623574145"
> print(paste("Precision for class 1: ", precision[2]))
[1] "Precision for class 1:  0.903225806451613"
```

Results and Conclusion

In this study, we employed various predictive modeling approaches including logistic regression, decision tree, Naïve Bayes, random forests, and k-nearest neighbors to analyze the Titanic dataset. The primary goal was to predict survival based on demographic and socio-economic factors such as age, gender, class, and fare, among others.

The models demonstrated varying degrees of accuracy, with the decision tree achieving the highest accuracy at 78.47%, followed by random forests at 77.75%, logistic regression at 76.32%, Naïve Bayes at 74.88%, and k-nearest neighbors at 66.03%. These results highlight the decision tree's superior performance in capturing the intricate patterns in the data, likely due to its ability to handle non-linear relationships and interactions between variables.

Key insights from this analysis include the significant impact of gender, age, and socio-economic status on survival chances. For instance, women and children in higher classes had a notably higher probability of survival. These findings align with historical accounts and provide a quantitative basis for understanding survival patterns during the Titanic disaster. The use of multiple models allowed for a comprehensive evaluation of different approaches, each offering unique strengths. For example, logistic regression provided a clear understanding of the individual impact of each variable, while the decision tree and random forests offered robustness and better handling of complex interactions.
In conclusion, this research not only sheds light on historical survival patterns but also demonstrates the effectiveness of various machine learning techniques in predictive modelling. These insights can inform strategies for managing human safety in future crises, emphasizing the importance of demographic and socio-economic factors in emergency planning.

References

1. Dimitri, G. M. (2023-2024) : Business Intelligence . DIISM, Universita' degli Studi di Siena.
2.Kaggle. (n.d.). Titanic - Machine Learning from Disaster. Retrieved from https://www.kaggle.com/c/titanic
3.Breiman, L., Friedman, J. H., Olshen, R. A., & Stone, C. J. (1984). Classification and Regression Trees. Wadsworth.
4.Quinlan, J. R. (1986). Induction of Decision Trees. Machine Learning, 1(1), 81-106.
5.R Documentation. (n.d.). rpart.plot: Plot an rpart model. Retrieved from https://www.rdocumentation.org/packages/rpart.plot/versions/3.0.9/topics/rpart.plot

By integrating these references and summarizing the key findings and methodologies, the report effectively communicates the results and significance of the study on predicting survival rates among Titanic passengers using machine learning techniques.